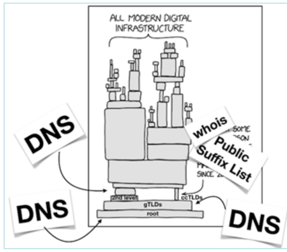
Domain Name System



Hostnames vs IP addresses

Two identifiers (labels) associated with any device connected to the Internet:

- Name, i.e., an alphanumeric label
  - variable length
  - variable structure (conform to organizational structure)
  - easy to remember
  - difficult to process by network devices (e.g., routers)

- IP address, i.e., a numeric label
  - fixed length (i.e., 32 or 128 bits)
  - fixed structure (conform to network structure)
  - difficult to remember
  - easy to process by network devices

Example: **`poldo.unipv.it`** vs **`193.204.34.3`**

Human vs machine perspectives

Historical perspective

At the time of ARPANET, all hostname & IP address pairs were stored in a text file called **hosts** (in the directory **/etc**) managed by SRI Network Information Center

- Changes were submitted to SRI by email
- Network administrators periodically downloaded (via **ftp**) new versions of the **hosts** file from SRI
- Network administrators could choose any hostname

The rapid growth of the Internet caused very serious management issues

- *Scalability:* SRI could not handle the load and traffic
- *Ambiguity*: SRI was unable to enforce unique hostnames
- *Inconsistency:* many machines had stale copies of the **hosts** file

The file **hosts** still exists and is used nowadays as a *static* table: **/etc/hosts** under Linux and **C:\Windows\System32\drivers\etc\hosts.txt** under Windows operating systems

Example

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97     rhino.acme.com          # source server
#       38.25.63.10     x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost
```

Simple solution

A fully centralized solution with a single database

- Advantage:
    - Simple to implement
- Disadvantages:
    - Reliability (i.e., single point of failure)
    - Performance/efficiency (i.e., traffic, latency)
    - Management/update
    - Scalability

The database became the bottleneck of the entire network

Problem & solution

The *problem* is how to obtain the *IP address* associated with a given *hostname*

The *solution* consists in the implementation of a specific service dedicated to the *translation* of hostnames into IP addresses

This service is the *Domain Name System*

Domain Name System: basic concepts

Domain Name System (DNS) is a friendly technology used to locate resources on the Internet

Invented in 1983 by Paul Mockapetris (RFCs 882 and 883 now obsolete)

DNS addresses two main problems:

- How names are mapped into IP addresses
- How applications/services obtain these mappings

Example:

$\texttt{https://}\underbrace{\texttt{www.google.com}}_{\text{hostname}}$

From the hostname applications need to derive the corresponding IP address (*lookup phase*)

Completely transparent

What DNS is about

It is defined as "*a general purpose naming service*" [RFCs 1034 "Domain Names – Concepts and Facilities" and 1035 "Domain Names – Implementation and Specification" by Paul Mockapetris published in 1987]

DNS is the core for all network applications/services (*critical infrastructure*)

It consists of two main components:

1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
2. an application layer *protocol* based on a *client/server model* over *UDP* protocol (on port 53) used to query the database

The basic service provided by DNS is the mapping (i.e., *translation*) between *hostnames* and *IP addresses*

Thinking about DNS

Humongous distributed database

- billion records, each simple

Handles many trillions of queries/day

- many more reads than writes
- performance matters since almost every Internet transaction interacts with DNS, so each msec counts

Organizationally, physically decentralized

- millions of different organizations responsible for their records

Reliability and security assurance